

AEM Release Notes – Release Version 2.6.0

Overview

This release introduces updates to improve ionic liquid modeling accuracy, expand report outputs, and enhance documentation for the AEM user base.

CLI Version: 2.26.1

Model Updates

- Improved treatment of ionic liquid cations in ACCC-based simulations
- Removed diameter-based residence time scaling for non-spherical cations
- ACCC-defined cation diameter is now used directly for applicable systems

Impact:

- Improved agreement with expected physical behavior and benchmark data
- More accurate conductivity predictions
- Enhanced modeling of ionic liquid and supercapacitor electrolytes

AEM Output Report Updates

- **Report 3 – Ion Solvation & Permittivity**
 - Added polarizability outputs:
 - Solution basis (Modified Clausius–Mossotti)
 - Solvent basis (Modified Clausius–Mossotti)
 - Expanded solvation energy and permittivity reporting
- **Report 6 – Activation Energies & Transport**
 - Added:
 - Conductivity-related activation energies
 - Net solvent–cation binding energy
 - Net cation desolvation energy
 - Relative permittivity (solution and solvent)
- **Report 7 – Large-Scale Simulation Optimization**
 - Now includes 14 optimization parameters (up from 8)
- **Report 17 – Electrolyte Ingress**
 - Added structured description for pore-filling dynamics vs. time across pore sizes
- **Report 19 – Cation Desolvation**
 - Added structured outputs for:
 - Ligand-wise binding energies (BE1–BE6)
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- Desolvation times (dt1–dt6)
 - $t\text{-}\lambda$ (desolvation favorability parameter)
 - Includes communal solvation (CS) effects
- **Report 20 – Communal Solvation (CS)**
 - Added reporting for:
 - Salt and solvent partitioning (bulk, CS, non-CS)
 - Solvation numbers and binding energies by region
- **Report 21 – DFN Transport Properties (New)**
 - Added outputs formatted for DFN modeling:
 - Diffusivity (D, D_{app})
 - Conductivity
 - Transference number (t^+)
 - Thermodynamic factor (TDF)
 - Activity coefficients (molal and molar basis)
 - Includes ion association and communal solvation corrections

Compound Library Updates

- **New Solvents:**
 - Methyl trifluoroethyl carbonate (FEMC)
 - Water (trace modeling capability)
 - Tetraethyl orthosilicate (TEOS)
- **New Salts:**
 - LiClO₄
 - TEABF₄
 - DMPBF₄

Documentation Updates

- Updated report descriptions into structured table format (Appendix)
- Added new report descriptions for Reports 17–21

Release Status

This is an interim release intended to support:

- Ionic liquid and supercapacitor electrolyte modeling
- Ongoing validation and benchmarking activities
- GUI integration and testing workflows

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